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Anthocyanins are coloured pigments found in the cell vacuole of some plant cells. Anthocyanins cannot move across undamaged cell membranes.

A student investigated how to extract anthocyanins from blueberries.

She mixed 10 g of crushed, fresh blueberries with 100 cm³ of extraction solvent for 1 hour.

She investigated three different extraction solvents:

- **E** – Ethanol, water and acid
- **F** – Ethanol and water
- **G** – Water

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When making up extraction solvent **E**, the student used a volume ratio of 70:30:1 ethanol:water:acid.

Tick (✓) **one** box that shows the most appropriate volumes she would use to make up 100 cm³ of extraction solvent **E**.

[1 mark]

63.6 cm³ ethanol, 27.3 cm³ water, 9.1 cm³ acid

☐

69.3 cm³ ethanol, 29.7 cm³ water, 1.0 cm³ acid

☐

70.0 cm³ ethanol, 30.0 cm³ water, 1.0 cm³ acid

☐

70.7 cm³ ethanol, 30.3 cm³ water, 1.0 cm³ acid

☐

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The student kept constant:

- the mass of fresh blueberries
- the volume of extraction solvent
- the time for the mixture to stand.

Name **two** other variables the student should have kept constant during this investigation.

[2 marks]

1 _____

2 _____

Her results are shown in **Figure 4**.

A bar chart showing the absorbance of three extraction solvents, E, F, and G. The y-axis is labeled 'Absorbance / arbitrary units' and ranges from 0.0 to 10.0 in increments of 2.0. The x-axis is labeled 'Extraction solvent' and has three categories: E, F, and G. The bars are gray with black outlines. Solvent E has the highest absorbance, followed by F, and G has the lowest.

Extraction solvent	Absorbance / arbitrary units
E	~8.3
F	~4.8
G	~0.2

Use your knowledge of membrane structure to explain the results in **Figure 4**.
[4 marks]

[illegible]

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A different student did this investigation. He did **not** have a colorimeter.

Describe a method this student could use to prepare colour standards and use them to give data for the total anthocyanin extracted.

[3 marks]

[illegible]

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ANSWER IN THE SPACES PROVIDED**

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